

SOFTWARE REQUIREMENTS SPECIFICATION

# Smart Hospital Management System

Role-based hospital operations platform with an AI copilot

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| Document           | Software Requirements Specification (SRS)                                                                             |
| Product            | Smart Hospital Management System (SHMS)                                                                               |
| Version            | 1.0 (Initial release)                                                                                                 |
| Date               | 19 September 2026                                                                                                     |
| Prepared by        | Ishan Singh                                                                                                           |
| Document structure | Based on IEEE Std 830-1998 / ISO/IEC/IEEE 29148                                                                       |
| Technology         | React 18 + TypeScript + Tailwind CSS   Node.js + Express   Prisma ORM   PostgreSQL / SQLite   Google Gemini 3.6 Flash |

## Revision History

| Version | Date        | Author      | Description                                                                  |
|---------|-------------|-------------|------------------------------------------------------------------------------|
| 1.0     | 19 Sep 2026 | Ishan Singh | Initial SRS covering all nine roles, AI copilot, architecture, RBAC and NFRs |

## Table of Contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>1. Introduction.....</b>                                     | <b>3</b>  |
| <b>2. Overall Description .....</b>                             | <b>3</b>  |
| <b>3. System Architecture .....</b>                             | <b>5</b>  |
| <b>4. Functional Requirements .....</b>                         | <b>6</b>  |
| 4.1 Authentication and Session Management .....                 | 6         |
| 4.2 Doctor Role.....                                            | 6         |
| 4.3 Nurse Role.....                                             | 7         |
| 4.4 Pharmacy Role .....                                         | 7         |
| 4.5 Laboratory Role .....                                       | 7         |
| 4.6 Finance Role .....                                          | 8         |
| 4.7 HRMS Role.....                                              | 8         |
| 4.8 Mess / Kitchen Role.....                                    | 8         |
| 4.9 Admin Role .....                                            | 9         |
| 4.10 Patient Portal .....                                       | 9         |
| 4.11 AI Copilot Module (/ai) .....                              | 9         |
| 4.12 IPD, Audit and Notification Services (Cross-cutting) ..... | 10        |
| <b>5. End-to-End Workflows.....</b>                             | <b>10</b> |
| <b>6. Role-Based Access Control Matrix .....</b>                | <b>11</b> |
| <b>7. Data Requirements.....</b>                                | <b>11</b> |
| <b>8. External Interface Requirements.....</b>                  | <b>12</b> |
| <b>9. Non-Functional Requirements.....</b>                      | <b>13</b> |
| <b>10. Constraints and Future Enhancements.....</b>             | <b>13</b> |

## 1. Introduction

### 1.1 Purpose

This document specifies the functional and non-functional requirements of the **Smart Hospital Management System (SHMS)**, a web-based platform that digitises day-to-day hospital operations. It is the agreed baseline for design, implementation, testing and evaluation. Intended readers are developers, testers, project supervisors and hospital stakeholders.

### 1.2 Scope

SHMS replaces disconnected paper and spreadsheet workflows with a single role-based system covering outpatient (OPD) consultations, inpatient (IPD) care, nursing, pharmacy, laboratory, billing, HR and payroll, diet and kitchen management, and patient self-service. A role-scoped AI copilot powered by Google Gemini produces on-demand briefings and answers queries using live hospital data.

**Out of scope (v1.0):** insurance claim adjudication or TPA integration, PACS/DICOM image storage, biomedical device integration, telemedicine video calls, and native mobile applications.

### 1.3 Definitions and Abbreviations

| Term         | Meaning                                             | Term | Meaning                           |
|--------------|-----------------------------------------------------|------|-----------------------------------|
| OPD / IPD    | Outpatient / Inpatient department                   | MAR  | Medication Administration Record  |
| UHID         | Unique Hospital ID of a patient                     | HRMS | Human Resource Management System  |
| EHR          | Electronic Health Record                            | RBAC | Role-Based Access Control         |
| ICD-10       | International Classification of Diseases, 10th rev. | JWT  | JSON Web Token                    |
| CL / ML / EL | Casual / Medical (Sick) / Earned Leave              | ORM  | Object-Relational Mapper (Prisma) |
| NPO          | Nil per os (no oral intake) diet order              | LLM  | Large Language Model (Gemini)     |
| PSI          | Pressure unit used for oxygen cylinders             | SPA  | Single-Page Application           |

### 1.4 References

- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018 - guidance on SRS content and structure.
- SHMS role-wise feature documentation, RBAC permission matrix and System Architecture (Level 1) diagrams supplied with the project.

### 1.5 Document Overview

Section 2 describes the product and its users. Section 3 presents the architecture. Section 4 lists detailed functional requirements per role, Section 5 shows end-to-end workflows and Section 6 the access-control matrix. Sections 7-10 cover data, interfaces, quality attributes, constraints and future scope. Requirement IDs (e.g. FR-DOC-03) are unique and traceable.

## 2. Overall Description

### 2.1 Product Perspective

SHMS is a self-contained, three-tier web application: a React single-page client, a Node.js/Express REST API, and a relational database accessed through Prisma. It integrates with the Google Gemini REST API for AI features and, optionally, with SMTP and SMS gateways and file storage for notifications and documents.

### 2.2 Product Functions

- **Clinical:** OPD queue and consultation studio, EHR history, ICD-10 diagnosis, prescriptions, diagnostic orders, encounters, IPD admission and bed allocation.
- **Care delivery:** vitals telemetry, MAR, nursing notes and handover, oxygen cylinder monitoring, inpatient diet prescription and kitchen fulfilment.
- **Support services:** pharmacy dispensing with batch/expiry stock control, laboratory order-to-report lifecycle.
- **Business:** unbilled-charge consolidation, invoicing, discounts, multi-mode payments, HR onboarding, attendance, leave approval and payroll.

- **Governance and insight:** executive KPIs, user administration, immutable audit logs, and role-scoped Gemini briefings and patient assistant.

## 2.3 User Classes and Characteristics

| Role                  | Primary responsibilities                                                   | Home route |
|-----------------------|----------------------------------------------------------------------------|------------|
| <b>Admin</b>          | Operations control, KPIs, user accounts, audit logs, full system oversight | /admin     |
| <b>Doctor</b>         | OPD consultations, diagnosis, prescriptions, lab orders, diet prescribing  | /doctor    |
| <b>Nurse</b>          | Ward roster, vitals, MAR, nursing notes, oxygen tracking                   | /nurse     |
| <b>Pharmacy</b>       | Prescription dispensing, medicine stock and batch management               | /pharmacy  |
| <b>Laboratory</b>     | Sample collection, test processing, result entry and report verification   | /lab       |
| <b>Finance</b>        | Charge consolidation, invoices, discounts, payments, revenue audit         | /finance   |
| <b>HRMS</b>           | Staff onboarding, leave approval, attendance, payroll                      | /hrms      |
| <b>Mess / Kitchen</b> | Meal preparation and delivery for admitted inpatients                      | /mess      |
| <b>Patient</b>        | Self-service appointments, prescriptions, reports, bills, AI assistant     | /portal    |

## 2.4 Operating Environment and Technology Stack

| Layer          | Technology                                                                                                                                                  |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Client</b>  | React 18, TypeScript, Vite, Tailwind CSS, React Router v6 (role-protected routes), Axios, Recharts; modern evergreen browsers on desktop, tablet and mobile |
| <b>API</b>     | Node.js, Express, TypeScript; CORS and Helmet; JWT authentication; RBAC middleware; audit logging                                                           |
| <b>Data</b>    | Prisma ORM; PostgreSQL (production) with SQLite for local development                                                                                       |
| <b>AI</b>      | Google Gemini 3.6 Flash via REST, invoked from a dedicated telemetry service                                                                                |
| <b>Tooling</b> | Git for version control; HTTP/REST (JSON) between client and API                                                                                            |

## 2.5 Design and Implementation Constraints

- Every API endpoint must enforce authentication and role authorization on the server; UI hiding alone is not sufficient.
- Multi-step operations that change stock, beds or money (dispensing, bed allocation, payment) must run as atomic database transactions.
- AI calls are made only on explicit user action and only with data the caller's role may already access.

## 2.6 Assumptions and Dependencies

- Users have reliable network access and a modern browser; staff accounts are created by HRMS or Admin.
- A valid Gemini API key and outbound internet access are available for AI features; other modules work without them.
- Pricing values (e.g. OPD fee of INR 800, lab test prices) are configured by the hospital and stored as master data.

### 3. System Architecture

#### 3.1 Architectural Overview

SHMS follows a layered architecture. Requests flow from the role-aware React client through a secured Express gateway into independent business-logic services, which use Prisma to reach the database and a dedicated service to reach Gemini. Figure 1 shows the components.

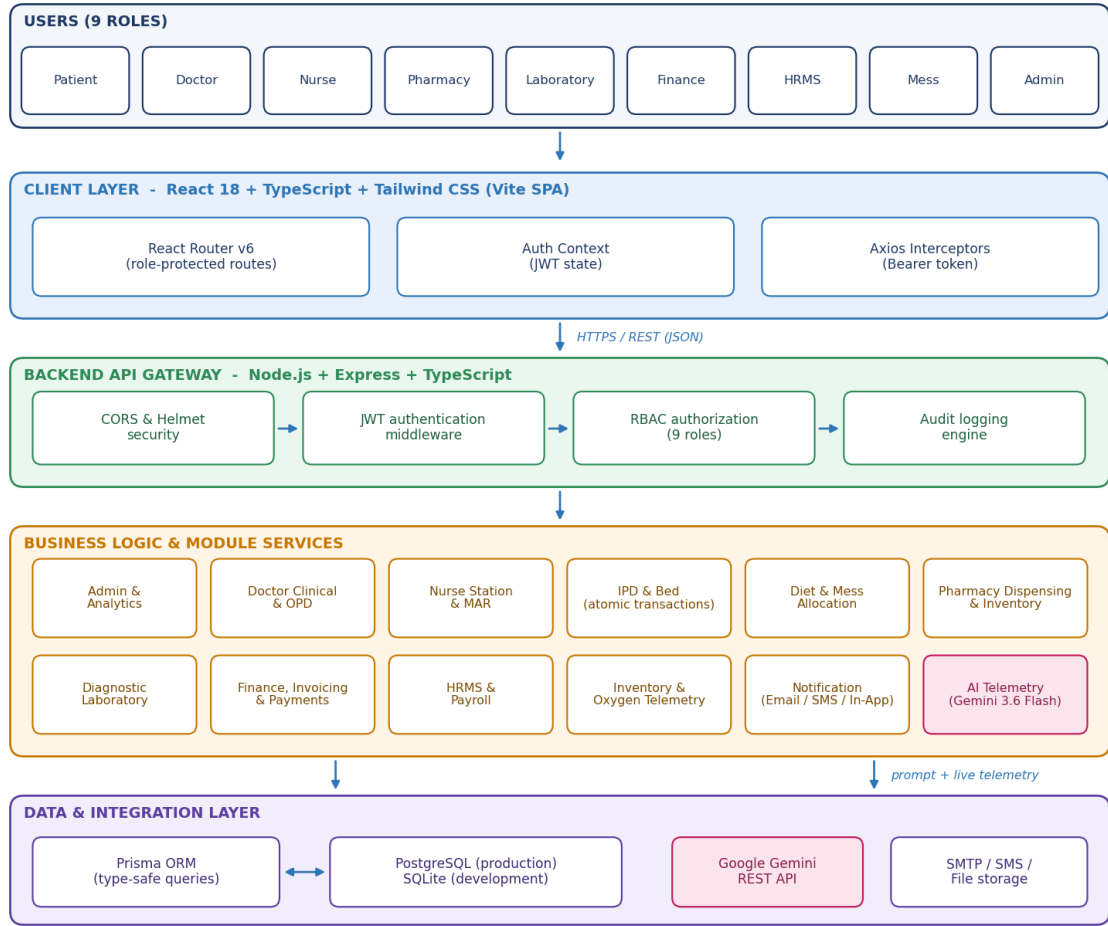


Figure 1 - SHMS layered system architecture

#### 3.2 Layer Responsibilities

| Layer       | Components                                                                                                    | Responsibility                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Client      | React SPA, Auth Context, React Router v6, Axios interceptors                                                  | Render role-specific dashboards, keep JWT state, block unauthorised routes, attach Bearer token to every request |
| API gateway | CORS and Helmet, JWT middleware, RBAC (9 roles), audit engine                                                 | Harden HTTP, verify identity, authorise by role and resource, record every significant action                    |
| Services    | Admin, Doctor/OPD, Nurse/MAR, IPD/Bed, Diet, Pharmacy, Lab, Finance, HRMS, Inventory/Oxygen, Notification, AI | Encapsulate module business rules; each service owns its data access through Prisma                              |
| Data / AI   | Prisma ORM, PostgreSQL/SQLite, Gemini REST API, SMTP/SMS                                                      | Persist data with ACID guarantees; supply prompt plus live telemetry to Gemini and return summaries              |

### 3.3 Request Lifecycle

1. The user logs in; the API returns a signed JWT that the client stores in Auth Context.
2. Axios attaches the Bearer token; React Router allows only routes permitted for the user's role.
3. The gateway validates CORS/Helmet rules, verifies the JWT and checks RBAC for the requested resource.
4. The matching service executes business logic, using Prisma transactions where required.
5. The audit engine records user, role, action, module, IP address and timestamp; the JSON response is returned.

## 4. Functional Requirements

Requirement keywords: **shall** = mandatory. Each table lists the feature (with its UI route) and the requirement.

### 4.1 Authentication and Session Management

| ID         | Feature / Route   | Requirement                                                                                                     |
|------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| FR-AUTH-01 | Login             | The system shall authenticate users by official email and password and issue a signed JWT on success.           |
| FR-AUTH-02 | Role routing      | After login the client shall redirect the user to the home route of their role and block all other role routes. |
| FR-AUTH-03 | Account status    | Accounts marked Inactive by Admin shall be refused login and existing tokens shall be rejected.                 |
| FR-AUTH-04 | Credential safety | Passwords shall be stored only as salted hashes; the API shall never return them.                               |

### 4.2 Doctor Role

*Primary purpose: OPD consultations, clinical encounters, diagnostic ordering, prescription building, inpatient diet prescribing and leave requests.*

| ID        | Feature / Route                           | Requirement                                                                                                                                                                                                                                  |
|-----------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-DOC-01 | OPD Queue<br>/doctor                      | Display a live patient token queue with status badges SCHEDULED, IN_PROGRESS and COMPLETED.                                                                                                                                                  |
| FR-DOC-02 | Consultation Studio                       | Record symptoms, physical examination notes, ICD-10 diagnosis codes (e.g. I10 - Essential Hypertension), multi-drug prescription line items and diagnostic lab orders in one encounter.                                                      |
| FR-DOC-03 | Consultation charge                       | On completing a consultation the system shall automatically generate an OPD consultation charge of INR 800 as an unbilled charge.                                                                                                            |
| FR-DOC-04 | Patients Directory<br>/doctor/patients    | Provide a patient list searchable by name, UHID and phone, showing demographics, blood group and allergy alerts.                                                                                                                             |
| FR-DOC-05 | EHR History Viewer                        | Show previous visits, past diagnoses and a medical timeline for the selected patient.                                                                                                                                                        |
| FR-DOC-06 | Clinical Encounters<br>/doctor/encounters | Keep a historical repository of signed encounter notes and treatment summaries.                                                                                                                                                              |
| FR-DOC-07 | Prescriptions<br>/doctor/prescriptions    | List issued prescriptions with medicine, dosage, frequency and fulfilment status (PENDING, DISPENSED); allow direct issue through a modal without opening the Consultation Studio.                                                           |
| FR-DOC-08 | Lab Orders<br>/doctor/lab-orders          | Track Pathology, Biochemistry and Radiology orders through ORDERED, SAMPLE_COLLECTED, PROCESSING and COMPLETED.                                                                                                                              |
| FR-DOC-09 | Inpatient Diet<br>/diet                   | Show the inpatient census (ward, bed, allergy alerts); assign or edit diet type (NORMAL, DIABETIC, LOW_SODIUM, HIGH_PROTEIN, RENAL, SOFT_DIET, LIQUID_DIET, NPO) and customise Breakfast, Lunch and Dinner menus with clinical instructions. |
| FR-DOC-10 | Leave Portal<br>/doctor/leave             | Apply for CL, ML or EL and view live leave balance and HRMS approval status.                                                                                                                                                                 |
| FR-DOC-11 | AI Briefing                               | Generate on demand a briefing of clinical census, active inpatients, risk/allergy flags and pending urgent lab orders.                                                                                                                       |

### 4.3 Nurse Role

*Primary purpose: inpatient ward care, vitals monitoring, MAR, nursing notes, oxygen tracking and diet-plan viewing.*

| ID        | Feature / Route                       | Requirement                                                                                                                                   |
|-----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| FR-NUR-01 | Ward Roster<br>/nurse                 | List active admitted patients with bed number, admission reason and assigned doctor.                                                          |
| FR-NUR-02 | Vitals Logger<br>/nurse/vitals        | Record blood pressure, pulse (BPM), temperature (F), SpO2 (%), respiratory rate and weight; show a historical telemetry timeline per patient. |
| FR-NUR-03 | MAR<br>/nurse/medications             | Show the real-time schedule of prescribed drugs and log each administration with dose, route (Oral, IV, IM), timestamp and nurse remarks.     |
| FR-NUR-04 | Nursing Notes<br>/nurse/notes         | Create progress notes, shift handover reports and clinical incident logs.                                                                     |
| FR-NUR-05 | Oxygen Telemetry<br>/inventory/oxygen | Monitor central-store and ward cylinders (PSI, capacity) and raise refill alerts.                                                             |
| FR-NUR-06 | Diet View<br>/diet                    | Provide read-only access to admitted patients' meal requirements and restrictions.                                                            |
| FR-NUR-07 | Leave Portal<br>/nurse/leave          | Submit leave requests that are automatically routed to HRMS.                                                                                  |

### 4.4 Pharmacy Role

*Primary purpose: prescription dispensing, drug stock inventory, batch expiry management and dispensing charges.*

| ID        | Feature / Route                        | Requirement                                                                                                                                                                                                                          |
|-----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-PHR-01 | Dispensing Queue<br>/pharmacy          | Show live incoming prescriptions issued by doctors.                                                                                                                                                                                  |
| FR-PHR-02 | Atomic Dispense                        | In a single transaction the system shall verify current stock, deduct inventory, create the dispensing record, generate the pharmacy charge and mark the prescription DISPENSED; on any failure the whole operation shall roll back. |
| FR-PHR-03 | Stock Directory<br>/pharmacy/inventory | Maintain the medicine catalogue (generic name, category, dosage form, unit price, total stock, reorder threshold) and alert on low-stock and near-expiry items.                                                                      |
| FR-PHR-04 | Batch Registry<br>/pharmacy/batches    | Record batch number, manufacturing and expiry dates, purchase cost and selling rate.                                                                                                                                                 |
| FR-PHR-05 | Leave Portal<br>/pharmacy/leave        | Allow pharmacists to apply for leave.                                                                                                                                                                                                |

### 4.5 Laboratory Role

*Primary purpose: pathology and radiology orders, sample collection, processing and verified report publishing.*

| ID        | Feature / Route            | Requirement                                                                                                                                     |
|-----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-LAB-01 | Test Queue<br>/lab         | Show incoming lab orders from OPD consultations and IPD wards.                                                                                  |
| FR-LAB-02 | Lifecycle actions          | Support the actions Collect Sample, Process Test, then Enter Results and Publish, updating the order status at each step.                       |
| FR-LAB-03 | Result Entry Studio        | Capture quantitative or qualitative values, normal biological range, units and an abnormality flag (Normal / Abnormal).                         |
| FR-LAB-04 | Verification               | Require formal verification and signing before a report becomes visible to the doctor and patient.                                              |
| FR-LAB-05 | Test Catalog<br>/lab/tests | Maintain investigations (e.g. CBC, Lipid Profile, HbA1c, LFT, KFT, Chest X-Ray) with standard pricing; completed tests post an unbilled charge. |
| FR-LAB-06 | Leave Portal<br>/lab/leave | Allow lab technicians to apply for leave.                                                                                                       |

## 4.6 Finance Role

*Primary purpose: billing consolidation, discount waivers, invoicing, cashier payments and revenue audit.*

| ID        | Feature / Route                | Requirement                                                                                                                                                            |
|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-FIN-01 | Unbilled Charges /finance      | Aggregate unbilled charges from OPD consultations, pharmacy dispenses, lab tests and IPD daily bed rates.                                                              |
| FR-FIN-02 | Consolidation                  | Allow multi-select of charges to be consolidated into a single invoice.                                                                                                |
| FR-FIN-03 | Invoices Hub /finance/invoices | Show itemised invoices with subtotal, discount, tax, final amount, paid amount and balance.                                                                            |
| FR-FIN-04 | Discount Waiver                | Apply a flat (INR) or percentage (%) discount and log the reason.                                                                                                      |
| FR-FIN-05 | Record Payment                 | Accept payments by Cash, Credit/Debit Card, UPI or Insurance; generate a receipt number (RCP-xxxxx) and update paid amount and balance.                                |
| FR-FIN-06 | Payroll view                   | Provide read-only access to payroll records for financial reconciliation.                                                                                              |
| FR-FIN-07 | Leave / AI Briefing            | Provide /finance/leave; on demand generate a briefing of billed revenue, collected cash flow, collection rate (%), outstanding receivables and payment-mode breakdown. |

## 4.7 HRMS Role

*Primary purpose: workforce management, onboarding by role and department, leave approvals and payroll liability.*

| ID        | Feature / Route             | Requirement                                                                                                                                                                                                                                                                                                              |
|-----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-HRM-01 | Staff Directory /hrms       | Show the complete employee roster across all departments.                                                                                                                                                                                                                                                                |
| FR-HRM-02 | Onboarding modal            | Create staff by category (Doctor, Nurse, Pharmacist, Lab Tech, Finance, Mess, HRMS, Admin) capturing official email, login password, department, designation, salary structure, shift timing, medical council licence number and doctor specialisation; the linked user account shall be created with the matching role. |
| FR-HRM-03 | Leave Approval /hrms/leaves | Provide a unified inbox of leave requests from all staff with one-click Approve or Reject and audit remarks; balances update on approval.                                                                                                                                                                                |
| FR-HRM-04 | Attendance /hrms/attendance | Log daily check-in/check-out timestamps and shift status (PRESENT, ABSENT, ON_LEAVE).                                                                                                                                                                                                                                    |
| FR-HRM-05 | Payroll /hrms/payroll       | Maintain disbursement records (Basic, HRA, Allowances, Deductions, Net Pay) and offer one-click monthly payroll generation.                                                                                                                                                                                              |
| FR-HRM-06 | AI Briefing                 | On demand summarise total monthly salary liability (INR), department payroll breakdown and pending leave requests.                                                                                                                                                                                                       |

## 4.8 Mess / Kitchen Role

*Primary purpose: inpatient meal preparation, kitchen queue fulfilment and delivery.*

| ID        | Feature / Route        | Requirement                                                                                                                                          |
|-----------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-MES-01 | Kitchen Queue /mess    | Schedule Breakfast, Lunch and Dinner for all admitted inpatients, showing ward, bed number, meal type and scheduled time.                            |
| FR-MES-02 | Privacy isolation      | The API shall redact medical diagnoses, prescriptions and physician notes from every kitchen response (HIPAA-aligned isolation), not only in the UI. |
| FR-MES-03 | Fulfilment actions     | Provide Mark Prepared and Mark Delivered actions that progress each meal PENDING, PREPARED, DELIVERED.                                               |
| FR-MES-04 | Food inventory / Leave | Allow create/update of food stock items; provide /mess/leave.                                                                                        |

## 4.9 Admin Role

*Primary purpose: hospital control plane, executive analytics, account management and security auditing.*

| ID        | Feature / Route                 | Requirement                                                                                                                                                      |
|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-ADM-01 | KPI Dashboard<br>/admin         | Show real-time cards: Total Patients, Today's Consultations, Active Inpatients, Bed Occupancy %, Pending Lab Orders, Low-Stock Items, Today's and Total Revenue. |
| FR-ADM-02 | Charts                          | Render Weekly Appointment Trends and Departmental Revenue Streams as graphs.                                                                                     |
| FR-ADM-03 | User Management<br>/admin/users | List accounts of all 9 roles and toggle each between Active and Inactive.                                                                                        |
| FR-ADM-04 | Audit Logs<br>/admin/audit-logs | Present the immutable audit trail: user ID, role, action type, target module, IP address, timestamp and metadata.                                                |
| FR-ADM-05 | AI Briefing                     | On demand summarise operations, income streams by department, bed-capacity headroom and supply-chain warnings.                                                   |

## 4.10 Patient Portal

*Primary purpose: self-service access to personal medical history, digital reports and billing.*

| ID        | Feature / Route                        | Requirement                                                                                       |
|-----------|----------------------------------------|---------------------------------------------------------------------------------------------------|
| FR-PAT-01 | Care Dashboard<br>/portal              | Show health summary, UHID, blood group, recorded allergies and upcoming appointments.             |
| FR-PAT-02 | Appointments<br>/portal/appointments   | List consultation tokens with date, doctor and status; allow booking with department specialists. |
| FR-PAT-03 | Prescriptions<br>/portal/prescriptions | Show medicine, dosage, frequency (e.g. 1-0-1) and doctor instructions.                            |
| FR-PAT-04 | Lab Reports<br>/portal/reports         | View and download verified laboratory and radiology reports with result values and normal ranges. |
| FR-PAT-05 | Bills<br>/portal/bills                 | Show itemised invoices, discounts, paid amounts, balances and payment receipts.                   |
| FR-PAT-06 | Data isolation                         | A patient shall be able to access only their own records.                                         |

## 4.11 AI Copilot Module (/ai)

| ID       | Feature / Route    | Requirement                                                                                                                                                                                |
|----------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AI-01 | Role-gated scoping | The backend shall build AI context only from telemetry that the logged-in role is permitted to read.                                                                                       |
| FR-AI-02 | Quick prompts      | Offer one-click prompt pills tailored for Admin, Doctor, HRMS, Finance and Patient workflows.                                                                                              |
| FR-AI-03 | On-demand briefing | Generate markdown briefings only when the user clicks; no AI call shall be made on page load.                                                                                              |
| FR-AI-04 | Patient assistant  | Answer questions on appointment schedules, active medications, lab-test explanations and hospital timings, always with an emergency safety disclaimer; it shall not diagnose or prescribe. |
| FR-AI-05 | Fault tolerance    | If Gemini is unreachable, show a clear message and keep all other modules functional.                                                                                                      |

## 4.12 IPD, Audit and Notification Services (Cross-cutting)

| ID        | Feature / Route     | Requirement                                                                                                                                                        |
|-----------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-IPD-01 | Admission & beds    | Doctors shall create and update admissions; nurses see them read-only. Bed allocation and discharge shall be atomic so a bed can never be double-assigned.         |
| FR-IPD-02 | Discharge & charges | Discharge shall free the bed and stop accrual of daily bed-rate charges to the unbilled charge pool.                                                               |
| FR-SYS-01 | Audit logging       | Every authenticated action that reads sensitive data or changes state shall create an append-only audit entry.                                                     |
| FR-SYS-02 | Notifications       | The Notification Service shall send email, SMS and in-app alerts (appointment confirmation, report ready, leave decision, low stock). Planned for v1.0 - optional. |

## 5. End-to-End Workflows

| Workflow              | Sequence                                                                                                                                                                                                                                                                                                            |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OPD to bill</b>    | Patient books appointment; token appears in doctor queue; doctor consults, prescribes and orders labs (INR 800 charge auto-generated); pharmacy dispenses and lab publishes (each posts a charge); Finance consolidates charges into an invoice, applies discount, records payment; patient views bill and receipt. |
| <b>Prescription</b>   | Doctor issues prescription (PENDING); Pharmacy queue receives it; atomic dispense deducts stock and posts charge; status becomes DISPENSED; patient sees it in the portal.                                                                                                                                          |
| <b>Lab order</b>      | Doctor orders test (ORDERED); technician collects sample, processes, enters results and verifies; report published; visible to doctor and patient.                                                                                                                                                                  |
| <b>Inpatient care</b> | Doctor admits patient and allocates bed; doctor prescribes diet; kitchen prepares and delivers meals; nurse records vitals and MAR; discharge frees bed and closes IPD charges.                                                                                                                                     |
| <b>Leave</b>          | Staff applies in the role's leave portal; request routes to HRMS inbox; HRMS approves or rejects with remarks; balance and status update; action is audited.                                                                                                                                                        |

## 6. Role-Based Access Control Matrix

**Legend:** F = full access, C = create, R = read, U = update, own = only the user's own records, limited = restricted fields only, - = no access. The gateway shall enforce this matrix on every endpoint.

| Module               | Admin | Doctor  | Nurse     | Patient | Pharmacy  | Lab       | Finance   | HRMS | Mess        |
|----------------------|-------|---------|-----------|---------|-----------|-----------|-----------|------|-------------|
| User Management      | F     | -       | -         | -       | -         | -         | -         | -    | -           |
| Appointments         | F     | U       | R/U       | C/R     | -         | -         | R         | -    | -           |
| Patient Profile      | F     | R/U     | R         | R/U own | R limited | R limited | R limited | -    | R limited   |
| Diagnosis            | F     | C/U     | R         | R       | -         | -         | -         | -    | -           |
| Prescription         | F     | C/U     | R         | R       | R         | -         | -         | -    | -           |
| Medical Reports      | F     | C/U     | R limited | R own   | -         | C/U lab   | -         | -    | -           |
| Vitals               | F     | R       | C/U       | R own   | -         | -         | -         | -    | -           |
| Nursing Notes        | F     | R       | C/U       | -       | -         | -         | -         | -    | -           |
| Medication Admin.    | F     | R       | C/U       | R own   | R         | -         | -         | -    | -           |
| Admissions           | F     | C/U     | R         | R own   | -         | -         | R         | -    | -           |
| Bed Management       | F     | R       | R/U       | R own   | -         | -         | R         | -    | -           |
| Laboratory           | F     | C       | R         | R own   | -         | F         | R charges | -    | -           |
| Pharmacy             | F     | C       | R         | R own   | F         | -         | R charges | -    | -           |
| Inventory            | F     | -       | -         | -       | C/U       | C/U       | R         | -    | C/U food    |
| Oxygen               | F     | R       | C/U       | -       | -         | -         | R         | -    | -           |
| Diet Plans           | F     | C/U     | R         | R own   | -         | -         | -         | -    | R/F fulfil. |
| Billing              | F     | R       | R limited | R own   | C         | C         | F         | -    | C charges   |
| Discount             | F     | -       | -         | -       | -         | -         | F         | -    | -           |
| Payments             | F     | -       | -         | C/R own | -         | -         | F         | -    | -           |
| HRMS                 | F     | -       | -         | -       | -         | -         | -         | F    | -           |
| Attendance           | F     | -       | -         | -       | -         | -         | -         | F    | -           |
| Leave                | F     | C/R own | C/R own   | -       | C/R own   | C/R own   | C/R own   | F    | C/R own     |
| Payroll              | F     | -       | -         | -       | -         | -         | R         | F    | -           |
| Audit Logs           | F     | -       | -         | -       | -         | -         | -         | -    | -           |
| AI Patient Assistant | F     | -       | -         | F own   | -         | -         | -         | -    | -           |
| AI Admin Copilot     | F     | -       | -         | -       | -         | -         | -         | -    | -           |

Note: the AI module is available to all roles as role-scoped briefings (FR-AI-01), while the matrix shows the full-access AI tools (Patient Assistant, Admin Copilot).

## 7. Data Requirements

### 7.1 Logical Data Entities

| Entity                                 | Key attributes                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| User / Employee                        | email, password hash, role (9), status Active/Inactive; category, department, designation, salary structure, shift, licence number, specialisation |
| Patient                                | UHID (unique), name, demographics, phone, blood group, allergies                                                                                   |
| Appointment / Encounter                | token, patient, doctor, department, date, status; symptoms, exam notes, ICD-10 diagnosis, signed flag                                              |
| Prescription / Item                    | medicine, dosage, frequency (e.g. 1-0-1), instructions, status PENDING/DISPENSED                                                                   |
| LabTest / LabOrder / LabResult         | catalogue name, category, price; order status lifecycle; value, unit, normal range, abnormality flag, verified by                                  |
| Admission / Bed                        | ward, bed number, admission reason, assigned doctor, admit and discharge dates, occupancy                                                          |
| Vitals / MedicationAdmin / NursingNote | BP, pulse, temperature, SpO2, respiratory rate, weight; dose, route, timestamp, remarks; note type, text, author                                   |
| OxygenCylinder                         | location (central/ward), PSI, capacity, refill flag                                                                                                |
| Medicine / Batch / DispenseRecord      | generic name, category, form, unit price, stock, reorder level; batch no., mfg/expiry, costs; quantity, charge                                     |

| Entity                       | Key attributes                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| DietPlan / MealOrder         | diet type, Breakfast/Lunch/Dinner menu, instructions; meal type, scheduled time, status                                 |
| Charge / Invoice / Payment   | source (OPD, PHARMACY, LAB, IPD), amount, billed flag; subtotal, discount, tax, final, paid, balance; mode, receipt no. |
| Leave / Attendance / Payroll | type CL/ML/EL, dates, status, remarks; check-in/out, status; basic, HRA, allowances, deductions, net pay                |
| AuditLog                     | user ID, role, action type, target module, IP address, timestamp, metadata (append-only)                                |

## 7.2 Data Rules

- UHIDs, invoice numbers and receipt numbers (RCP-xxxxx) shall be unique; monetary values are stored in INR with two decimals.
- Status fields shall use fixed enumerations (appointment, prescription, lab order, meal, attendance, leave) and only permitted transitions shall be accepted.
- Referential integrity and ACID transactions are mandatory; audit records shall never be updated or deleted.

## 8. External Interface Requirements

### 8.1 User Interface

The UI shall be responsive (desktop, tablet, mobile), styled with Tailwind CSS, show only navigation permitted for the user's role, use colour-coded status badges, modals for quick actions (issue prescription, record payment, apply discount, onboard employee) and render AI output as formatted markdown.

### 8.2 Software and Communication Interfaces

| Interface                    | Protocol / Purpose                                                                     |
|------------------------------|----------------------------------------------------------------------------------------|
| Client to API                | HTTPS, REST, JSON; Authorization: Bearer JWT                                           |
| API to database              | Prisma ORM over PostgreSQL (production) or SQLite (development)                        |
| API to Gemini                | HTTPS REST call to Google Gemini 3.6 Flash with role-scoped prompt and live telemetry  |
| API to notifications / files | SMTP for email; SMS gateway (Twilio or MSG91); cloud or local file storage for reports |

## 9. Non-Functional Requirements

The figures below are proposed targets to be validated during testing.

| ID     | Category        | Requirement                                                                                                                                  |
|--------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Performance     | At least 95% of API calls shall respond within 2 s and dashboards shall load within 3 s with 100 concurrent users; AI briefings within 15 s. |
| NFR-02 | Security        | JWTs shall expire; Helmet and a restrictive CORS policy are mandatory; all input shall be validated; HTTPS shall be used in production.      |
| NFR-03 | Authorization   | RBAC shall be enforced server-side for all nine roles and at resource level for own-record access.                                           |
| NFR-04 | Privacy         | Patient data shall be exposed on a need-to-know basis; the kitchen role shall never receive clinical data.                                   |
| NFR-05 | Reliability     | Target availability of 99%; all financial and stock operations transactional; daily database backups.                                        |
| NFR-06 | Usability       | Common tasks (dispense, record payment, log vitals) shall need no more than three interactions; consistent labels and badges.                |
| NFR-07 | Maintainability | Modular services, TypeScript type safety end to end, and Prisma schema migrations for controlled changes.                                    |
| NFR-08 | Portability     | Same codebase runs on SQLite (development) and PostgreSQL (production); supports current Chrome, Edge, Firefox and Safari.                   |
| NFR-09 | Scalability     | A stateless API so multiple instances can run behind a load balancer.                                                                        |
| NFR-10 | Auditability    | Every state change and sensitive read shall be traceable to a user, time and IP address.                                                     |

## 10. Constraints and Future Enhancements

---

### 10.1 Known Limitations

- AI output is informational and does not replace clinical judgement; the patient assistant carries an emergency disclaimer.
- Insurance is a recorded payment mode only; there is no claim processing workflow in v1.0.

### 10.2 Future Enhancements

- Insurance/TPA claim management, appointment reminders through WhatsApp, and online payment gateway integration.
- PACS/DICOM viewer for radiology, wearable/biomedical device integration, and multi-branch hospital support.
- Native mobile apps, multilingual UI, and tool-calling AI agents that can draft orders for clinician approval.